



Prevalence and molecular characterization of β -lactamase producers and fluoroquinolone resistant clinical isolates from North East India

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ABSTRACT

Introduction: The rapid emergence and variations of antibiotic resistance among common gram negative bacteria cause a significant concern specially in India and all over the world because of high mortality and morbidity rates.

Methods: In our study, we screened 189 bacterial isolates from Assam Medical College & Hospital, Dibrugarh for antibiotic resistance pattern and tried to identify the resistant genes causing responsible for β -lactam and fluoroquinolones resistance.

Results: More than 80% and 45% strains were resistant to all the 3rd generation cephalosporins, fluoroquinolones respectively. Among the 3rd generation cephalosporin resistant strains, 38% and 24% isolates were only ESBL and MBL producers respectively and 11% were reported to have both ESBL and MBL genes. The ESBL positive isolates have shown the dominance of CTX-M3 gene. VIM-1 gene was mostly reported in MBL producers. Our study probably for the first time reporting SIM-1 and SPM-1 MBL gene from India. Mutations in QRDR is found to be the primary cause of fluoroquinolone resistance along with efflux pump and PMQR presence.

Conclusion: The study represents the first detailed study on antibiotic resistance from NE India this could help to take control measures for the emerging antibiotic resistance in hospital and community based infections in North East India.

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Introduction

Antimicrobial resistance in gram negative pathogens (GNP) is a rapidly emerging crisis in modern medical world [1]. Antibiotics play a vital role in the treatment and management of various bacterial infections, leading to a reduction in morbidity and mortality in both human and animal. The usage of antimicrobials in agriculture, veterinary and medical enterprises have direct or indirect effects on the development of Antimicrobial resistance (AMR) by bacteria associated with animals and plants which may enter the food chain through consumption of meat, fish, vegetables or some other food sources. UN Ad hoc Interagency Coordinating Group on

Antimicrobial Resistance warns that, drug resistant diseases can cause 10 million deaths each year by 2050 [2]. In gram negative pathogens, *Escherichia coli* and *Klebsiella pneumoniae* are the two main predominant pathogens frequently reported to be associated with hospital acquired infection and other common infections. Emergence of multi drug resistant *Acinetobacter baumannii*, *Pseudomonas aeruginosa* and other members of *Enterobacteriaceae* also pose a significant threat in treatment of common infections [3]. Irrespective of age and boundary, antibiotic resistance can affect people across the globe. It is estimated that 5.7 million people, most in low- and middle-income countries (LMICs), die each year from treatable infectious diseases [4].

To understand the current spread of multi drug resistant bacteria, World Health Organization and Centers for Disease Control and Prevention, enlisted few multi drug resistant bacteria as potential threat for the world and where in carbapenemase (Metallo β -lactamase) producing *Acinetobacter baumannii*, *Pseudomonas aeruginosa* and 3rd generation cephalosporin resistant (Extended spectrum β lactamase producing) *Enterobacteriaceae* are on the high list of multi drug resistant threat bacteria and has been categorized as “priority pathogens” by WHO [5,6].

In India, antibiotic consumption rate by human is high (12.9×10^9 units in 2010, with 23% increase every year). Currently, at least 700,000 people die each year due to drug-resistant diseases, including 230,000 people who die from multidrug-resistant tuberculosis (WHO, 2019). Studies have shown that in India mortality rate

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due to drug resistant gram negative bacterial infection is 17.7%, which is higher than death caused by gram positive bacterial infection (10.8%) [7,8]. But there are limited reports on the prevalence of various resistant gram negative pathogens associated with clinical infections from North Eastern part of India which is visited by several international tourists every year. Prevalence of drug resistance among the commonly isolated pathogens causing clinical infections is not explored adequately in North East India. The phenotypic and genetic characterization of 3rd generation cephalosporin and carbapenem resistance has also not been explored extensively [9,10].

In this study we made an attempt to understand the prevalence and molecular determinants of β -lactamase producers and fluoroquinolone resistant isolates from clinical samples in North East India.

Materials and methods

Samples

A total of 208 clinical samples like urine, blood, tracheal aspirate, stool, wound swab, pus and sputum were collected from the laboratory of the Microbiology Department of Assam Medical College & Hospital, Dibrugarh (one of the largest government-run medical college and hospital in Assam) during 1st October 2018 to 30th October 2019 and included in the study and cultured in the growth media. The isolates were collected with due permission from the hospital authority along with the signed consent from the patients. The isolates obtained following culture in suitable media were then transported to Indian Council of Medical Research-Regional Medical Research Centre, Bhubaneswar for further processing for the identification of causative agent, antimicrobial susceptibility test (AST) profiling and molecular characterization of MDR isolates.

The study was approved by the institution ethical committee of ICMR-Regional Medical Research Centre, Bhubaneswar vide letter No ICMR-RMRCB/IHEC-2019/034.

Microbiological examination

The samples were subjected to standard techniques for identification of the bacterial isolates using suitable growth media like MacConkey's Agar, Blood Agar Eosin Methylene Blue (EMB) Agar (*E. coli*, *K. pneumoniae*, *Proteus* sp. We have used), Hektoen Enteric (HEA) agar (*Salmonella* sp., *Shigella* sp.), Hi Chrome MM agar (*Citrobacter* sp.), Leeds Acinetobacter Agar base (*Acinetobacter baumannii*) and Mannitol Agar w/Prilion (*Proteus* sp.) [11,12]. After overnight incubation at 37 °C, a single colony was isolated from the selective media and grown in pure culture and subsequently identified using standard procedure in which the suspected colonies were subjected to biochemical and serological tests using group-specific antisera (Denka Seiken Co., Ltd., Tokyo, Japan). The panel of biochemical tests used were IMViC test, Mannitol Motility test, Catalase test, Oxidase test, Triple Sugar Iron (TSI) Agar, Urease test (Hi-Media, Mumbai) [11,12]. Wherever more than a single type of colony was observed, all the suspected colonies were subjected to isolation and identification.

Determination of antimicrobial susceptibility and minimum inhibitory concentration

The test was conducted by applying a bacterial inoculum according to the 0.5 McFarland standard on the surface of Mueller-Hinton agar (Merck, Germany) plates. McFarland turbidity standard of 0.5 provides an optical density comparable to the density of a bacterial suspension 1.5×10^8 colony forming units (CFU/mL). The identified gram negative bacteria were further tested for antimicrobial susceptibility using 22 antibiotics of different classes by Kirby Bauer's Disk diffusion method in accordance with Clinical and Laboratory Standard Institute (CLSI) guidelines [13] using commercially available disks (Hi-media, Mumbai). *Escherichia coli* ATCC 25922 and

Staphylococcus aureus ATCC 25923 strains was included in the assay as quality control strains. All plates were incubated at 35 °C for 24 h, and the zones of growth inhibition around each of the antibiotic disks are measured to the nearest millimeter. On each occasion, the zone sizes in the plates were examined by at least two individuals.

Multi-drug resistance was defined as resistant to 3 or more antimicrobial categories.

Similarly, minimum inhibitory concentration (MIC) for 3rd generation cephalosporins (cefotaxime, ceftazidime) and fluoroquinolone (ciprofloxacin, norfloxacin) was determined by placing E-test (Hi-media). The breakpoints of the minimum inhibitory concentration of the antibiotics were analysed by per CLSI guideline [13].

Detection of ESBL and MBL production

The isolates resistant to 3rd generation cephalosporin (cefotaxime, ceftazidime) and carbapenem (imipenem, meropenem) were further tested for the presence of β -lactamase production. In accordance to CLSI guidelines, double disk diffusion method was performed. Cefotaxime (CTX) and cefotaxime + clavulanic acid (CEC) were used for (Extended Spectrum β Lactamase) ESBL and imipenem (IPM), imipenem + EDTA (IE) for (Metallo β -lactamase) MBL detection [14,15].

Genotypic identification of beta-lactamase producing genes

Bacterial DNA was isolated by Phenol-chloroform-isoamylalcohol method [16]. Then bacterial DNA was preserved in (-20°C) for future use. Polymerase chain reaction (PCR) was performed for determining beta-lactamase producing genes using specific primer and reaction condition [17].

For determination of ESBL enzymes following primers were used (*blaTEM*, *blaSHV*, *blaCTX-M*, *blaCTX-M3*, *blaOXA-1*, *blaOXA-7*) and for MBL enzymes the primers are used (*IMP-1*, *VIM-1*, *SIM-1*, *SPM-1*, *GIM-1*, *NDM-1*, *OXA-48*) [18,19]. The detailed sequence of the primers and reaction conditions are listed below in the Table 1. The *blaCTX-M* genes detected in the strains were subsequently confirmed to be *blaCTX-M3* by partial DNA sequencing and sequence analysis using MEGA5.5.

Screening of fluoroquinolone resistant strains

The isolates that were resistant to all five fluoroquinolone antibiotic disks (Ciprofloxacin, Norfloxacin, Gatifloxacin, Ofloxacin, Levofloxacin) were further screened for presence of quinolone resistant determining regions (QRDR) mutations and plasmid mediated quinolone resistance (PMQR) determinants.

Efflux pump assay

The Fluoroquinolone resistant strains were tested for efflux pump activity by the addition of carbonyl cyanide *m*-chlorophenylhydrazine (CCCP) to Muller Hinton agar (MHA) at a concentration 20 μ g/mL (<MIC). The MHA plates with and without CCCP were tested for MIC of Ciprofloxacin. Those showing decrease of MIC with CCCP were tested as efflux pump positive [20].

Analysis of mutations in quinolone resistant determining regions (QRDRs) in fluoroquinolone resistant strains

Efflux pump positive isolates and other five randomly selected bacterial strains were checked for the role of mutation in Quinolone resistant determining regions (QRDRs) for fluoroquinolone resistance. Bacterial strains DNA were amplified by PCR using published primers for *gyrA*, *gyrB*, *parC* and *parE* and reaction conditions

Table 1

Details of the primers, annealing temperature, and amplicon size of the various polymerase chain reaction (PCR) amplifications.

Sl. No	Primer	Sequence (5'-3')	Annealing temp (°C)	Amplicon Size (bp)
1	NDM-1	ACC GCC TGG ACC GAT GAC CA GCC AAA GTT GGG CGC GGT TG	58	264
2	OXA-48	GCGTGGTTAAGGATGAACAC CATCAAGTTCAACCAACCG	56	438
3	IMP	GGA ATA GAG TGG CTT AAY TCT C CCA AAC YAC TAS GTT ATC T		188
4	VIM	GAT GGT GTT TGG TCG CAT A CGA ATG CGC AGC ACC AG		390
5	GIM	TCG ACA CAC CTT GGT CTG AA AAC TTC CAA CTT TGC CAT GC	52	477
6	SPM	AAA ATC TGG GTA CGC AAA CG ACA TTA TCC GCT GGA ACA GG		271
7	SIM	TAC AAG GGA TTC GGC ATC G TAA TGG CCT GTT CCC ATG TG		570
8	blaTEM	GAGTATTCAACATTTTCGT ACCAATGCTTAATCAGTGA		857
9	blaSHV	TCGCCTGTGATTATCTCCC CGCAGATAAATCACCACAATG		768
10	blaCTX-M3	AATCACTGCGTCAGTTAC TTTATCCCCACAACCCAG	50	701
11	blaCTX-M	CCGCRATATGRTTGGTGGTG SCSATGTGCCAGYACCAAGTAA		593
12	blaOXA1	GCAGCGCCAGTGCATCAAC CCGCATCAAATGCCATAAGTG		198
13	blaOXA7	AGTTCTCTGCCGAAGCC TCTCAACCAACCAACCC		591
14	gyr A	TACACCGGTCAACATTGAGG TTAATGATTGCCGCGTCGG		648
15	par C	GTCTGAACCTGGCCTGAATGC AGCAGCTCGGAATATTTGACAA	64	249
16	qnr A	ATTTCTCACGCCAGGATTG GATCGGCAAAGGTTAGGTCA		516
17	qnr S	GCAAGTTCATTGAACAGGGT TCTAAACCGTCGAGTTCGGCG		428
18	Qep A	AACTGCTTGAGCCCGTAGAT GTCTACGCCATGGACCTCAC		596
19	aac(6')-Ib-cr	TTGCGATGCTCTATGAGTGGCTA CTCGAATGCTTGGCGTGT	55	482

[21,22]. The amplified region was then sequenced using an automatic sequencer (ABI 3130; Applied Biosystems, Foster City, CA, USA) which were further analysed for detection of mutation using SeqScape v3 (ABI 3130; Applied Biosystems) [23].

Screening of PMQR determinant

Same isolates were tested for the presence of plasmid mediated quinolone resistant (PMQR) determinants, *qnrA*, *qnrB*, *qnrC*, *qnrS* and two additional PMQR genes like - *aac(6')-Ib-cr* and *qepA* were screened for the fluoroquinolone resistant strains using published primers and reaction condition [23].

Results

Patients and isolates

A total of 189 bacterial colonies were obtained from 208 samples collected from Assam Medical College, Dibrugarh by the department of Life Science, Dibrugarh University and sent to RMRC Bhubaneswar for identification and characterization. Out of these 189 colonies, 110 colonies were identified as gram negative bacteria of different genus and species. The details of the patients (age, sex, ward, complication, prescribed medicine) were collected (Table 2). Bacterial strains were recovered from blood (n = 28; 25%), urine (n = 35; 32%), stool (n = 17; 15%), tracheal aspirate (n = 11; 10%), pus (n = 7; 6%), wound swab (n = 6; 5%), sputum (n = 6; 5%) (Table 3). Among these 110 strains, 47 (43%) were *E. coli*, 25 (23%) were *K. pneumoniae*, 13 (12%) *A. baumannii*, 9 (8%) *P. aeruginosa*,

5(4.5%) both *Shigella* sp. and *Citrobacter* sp., 3(2.7%) *Enterobacter* sp., 2(1.8%) *Salmonella* sp. and 1 (0.9%) *Proteus* sp. *E. coli* is was one of the major isolate obtained from blood, urine, stool samples (Table 3).

Antibiotic sensitivity among the clinical isolates

Among 110 gram negative bacteria, a wide variety of multi drug resistance was observed. Among them, 21 (19.09%) isolates were extremely drug resistant (XDR) and 85 (77.2%) were multi drug resistant (MDR) as per standardized international terminology created by European Centre for Disease Control (ECDC) and Centre for Disease Control & Prevention (CDC), Atlanta [23]. Only 15 isolates (13.6%) were susceptible to most of the categories of antibiotics. More than 85% strains were resistant to all the 3rd generation cephalosporins (CTX, CAZ, CTR), 45% were resistant to fluoroquinolones (CIP, NOR, OFX, GAT, LEV), chloramphenicols (CHL) and tetracyclin (TET) and more than 30% were resistant to aminoglycosides (AMK, GEN). Imipenem (IPM), cefotaxime + clavulanic acid (CEC) and piperacillin + tazobactam (TZP) resistance was observed in 30% of the strains isolated (Table 4).

Detection of β -lactamase producers

Bacterial isolates which were resistant to 3rd generation cephalosporins had minimum inhibitory concentration (MIC) ranged between 128 to >4096 mg/L which is above the cutoff of CLSI guidelines (CLSI [13]). All the 89 bacterial strains resistant to 3rd generation cephalosporins were tested by double disk diffusion

Table 2
Patient details and the list of microorganisms isolated from 110 samples.

Strain No	Age	Gender	Ward	Primary diagnosis	Primarily Prescribed Antibiotics	Source	Bacteria
AMCH 1	1	M	OBN	Sepsis	Amoxycillin	Blood	<i>A. baumannii</i>
AMCH 2	27	F	OPD	UTI	Ciprofloxacin	Urine	<i>E. coli</i>
AMCH 3	20	M	OBN	Sepsis	Cefotaxime Injection	Blood	<i>A. baumannii</i>
AMCH 4	7	F	PNW	Sepsis	Cefotaxime Injection	Blood	<i>E. coli</i>
AMCH 5	1	F	NICU	Sepsis	Cefotaxime Injection	Blood	<i>K. pneumoniae</i>
AMCH 6	63	M	MU-VII	Pneumoniae	Amoxycylav + Azithromycin	Tracheal Aspirate	<i>A. baumannii</i>
AMCH 7	55	F	MU-I	CKD + UTI	Ceftriaxone	Urine	<i>K. pneumoniae</i>
AMCH 8	22	F	MU-II	Dysentery	Metronidazole	Stool	<i>E. coli</i>
AMCH 9	32	F	SV-V	Left Abdominal Pain	Amoxycillin	Urine	<i>A. baumannii</i>
AMCH 10	50	F	IN	UTI	Ciprofloxacin	Urine	<i>E. coli</i>
AMCH 11	65	M	MU-VII	Pneumoniae	Amoxycylav + Azithromycin	Tracheal Aspirate	<i>E. coli</i>
AMCH 13	26	F	IN	UTI	Ciprofloxacin	Urine	<i>Citrobacter sp.</i>
AMCH 14	48	M	Ortho-I	LRTI	Amoxycillin	Sputum	<i>E. coli</i>
AMCH 15	82	M	SV-V	Left Abdominal Pain	Amoxycillin	Urine	<i>Citrobacter sp.</i>
AMCH 16	38	F	GOPD-III	UTI	Ciprofloxacin	Urine	<i>K. pneumoniae</i>
AMCH 17	16	M	SV-IV	Surgery	Amoxycylav	Wound Swab	<i>P. aeruginosa</i>
AMCH 18	20	F	PNWF I	UTI	Ciprofloxacin	Urine	<i>E. coli</i>
AMCH 21	43	F	MU-I	UTI	Ciprofloxacin	Urine	<i>E. coli</i>
AMCH 23	62	M	OPD	CKD + UTI	Ceftriaxone	Urine	<i>K. pneumoniae</i>
AMCH 25	40	F	MU-VII	LRTI	Amoxycillin	Sputum	<i>K. pneumoniae</i>
AMCH 26	61	M	PAED	LRTI	Amoxycylav	Sputum	<i>E. coli</i>
AMCH 27	43	F	MU-I	UTI	Ciprofloxacin	Urine	<i>E. coli</i>
AMCH 28	57	M	MU-VII	Pneumoniae	Amoxycylav + Azithromycin	Tracheal Aspirate	<i>P. aeruginosa</i>
AMCH 29	68	F	MU-VII	Pneumoniae	Amoxycylav + Azithromycin	Tracheal Aspirate	<i>A. baumannii</i>
AMCH 30	45	F	MU-V	UTI	Ciprofloxacin	Urine	<i>E. coli</i>
AMCH 31	50	F	SOPD-I	UTI	Ciprofloxacin	Urine	<i>E. coli</i>
AMCH 32	58	F	Ortho-2	Surgery	Ceftriaxone	Wound Swab	<i>K. pneumoniae</i>
AMCH 33	70	M	MU-VI	Benign Prostatic Hyperplasia	Nitrofurantoin	Urine	<i>E. coli</i>
AMCH 34	10	M	PAED	LRTI	Amoxycylav	Sputum	<i>E. coli</i>
AMCH 35	1	F	NICU	Newborn Sepsis	Cefotaxime Injection	Blood	<i>A. baumannii</i>
AMCH 36	35	M	Plas/Surgery	Surgical Wound	Ceftriaxone	Wound Swab	<i>P. aeruginosa</i>
AMCH 38	11	F	SU-IV	Surgical Wound	Amoxycylav	Wound Swab	<i>P. aeruginosa</i>
AMCH 39	69	M	P-Cabin 39	UTI	Nitrofurantoin	Urine	<i>E. coli</i>
AMCH 41	80	M	MU-VII	Pneumonia	Amoxycylav	Tracheal Aspirate	<i>P. aeruginosa</i>
AMCH 42	22	M	MU-II	Dysentery	Metronidazole	Stool	<i>E. coli</i>
AMCH 44	1	F	OBN	Sepsis	Cefotaxime Injection	Blood	<i>K. pneumoniae</i>
AMCH 45	20	M	URO	UTI	Ciprofloxacin	Urine	<i>E. coli</i>
AMCH 46	46	M	MU-VII	Dysentery	Metronidazole	Stool	<i>E. coli</i>
AMCH 47	33	F	MU-II	SLE	Nitrofurantoin	Urine	<i>K. pneumoniae</i>
AMCH 48	1	M	OBN	Sepsis	Cefotaxime Injection	Blood	<i>K. pneumoniae</i>
AMCH 50	4	M	OBN	Sepsis	Cefotaxime Injection	Blood	<i>P. aeruginosa</i>
AMCH 53	32	M	MU-2	Diarrhoea	Ofloxacin	Stool	<i>E. coli</i>
AMCH 54	30	F	Gynae OPD	UTI	Ciprofloxacin	Urine	<i>E. coli</i>
AMCH 55	66	F	MU-3	Diarrhoea	Norfloxacin	Stool	<i>E. coli</i>
AMCH 56	40	M	MU-2	Left Abdominal Pain	Nitrofurantoin	Urine	<i>E. coli</i>
AMCH 57	1	M	OBN	Sepsis	Cefixime Injection	Blood	<i>Citrobacter sp.</i>
AMCH 59	25	F	MU-1	CKD	Ceftriaxone	Urine	<i>E. coli</i>
AMCH 60	2	F	NICU	Sepsis	Ceftriaxone	Blood	<i>A. baumannii</i>
AMCH 61	7	M	PNW	Sepsis	Cefotaxime Injection	Blood	<i>K. pneumoniae</i>
AMCH 62	3	M	OBN	Sepsis	Cefixime	Blood	<i>E. coli</i>
AMCH 64	60	M	SU-I	UTI	Ciprofloxacin	Urine	<i>K. pneumoniae</i>
AMCH 68	10	M	PAED	LRTI	Amoxycillin	Sputum	<i>K. pneumoniae</i>
AMCH 69	23	F	MU-I	Dysentery	Metronidazole	Stool	<i>E. coli</i>
AMCH 70	35	M	SU-IV	Surgery	Amoxycylav	Wound Swab	<i>E. coli</i>
AMCH 71	5	F	PAED	UTI	Amoxycylav	Urine	<i>E. coli</i>
AMCH 73	34	M	MU-I	Dysentery	Ofloxacin	Stool	<i>E. coli</i>
AMCH 74	2	F	OBN	Sepsis	Ceftazidime	Blood	<i>E. coli</i>
AMCH 78	1	M	OBN	Sepsis	Amoxycillin	Blood	<i>P. aeruginosa</i>
AMCH 80	44	F	OPD	Abdominal Pain	Ciprofloxacin	Urine	<i>E. coli</i>
AMCH 84	1	M	OBN	Sepsis	Cefotaxime Injection	Blood	<i>E. coli</i>
AMCH 85	7	M	OBN	Sepsis	Cefixime Injection	Blood	<i>E. coli</i>
AMCH 86	34	M	MU-II	CKD + UTI	Ceftriaxone	Urine	<i>K. pneumoniae</i>
AMCH 87	55	F	MU-VI	Dysentery	Metronidazole	Stool	<i>K. pneumoniae</i>
AMCH 89	17	M	Ortho-I	UTI	Ciprofloxacin	Urine	<i>K. pneumoniae</i>
AMCH 92	12	F	OBN	Sepsis	Amoxycylav	Blood	<i>E. coli</i>
AMCH 93	8	M	OBN	Sepsis	Ceftriaxone	Blood	<i>E. coli</i>
AMCH 94	16	M	MU-V	Pneumoniae	Amoxycillin + Azithromycin	Tracheal Aspirate	<i>K. pneumoniae</i>
AMCH 96	29	F	OPD	UTI	Ciprofloxacin	Urine	<i>K. pneumoniae</i>
AMCH 97	1	M	OBN	Newborn Sepsis	Ceftriaxone	Blood	<i>E. coli</i>
AMCH 99	67	M	MU-VI	Pneumoniae	Amoxycylav + Azithromycin	Tracheal Aspirate	<i>A. baumannii</i>
AMCH 104	1	F	OBN	Newborn Sepsis	Cefotaxime Injection	Blood	<i>E. coli</i>
AMCH 105	35	M	SU-I	Cellulitis (L)	Cefotaxime	Pus	<i>A. baumannii</i>
AMCH 106	24	F	OPD	Abdominal Pain	Ciprofloxacin	Urine	<i>E. coli</i>
AMCH 108	32	F	OPD	UTI	Cefuraxime	Urine	<i>E. coli</i>
AMCH119	60	F	MU-I	UTI	Nitrofurantoin	Urine	<i>Citrobacter sp.</i>

Table 2 (Continued)

Strain No	Age	Gender	Ward	Primary diagnosis	Primarily Prescribed Antibiotics	Source	Bacteria
AMCH124	75	F	OPD	Wound Infection	Amoxycillin	Pus	<i>Enterobacter sp.</i>
AMCH127	55	M	OPD	Dysentery	Ciprofloxacin	Stool	<i>Salmonella sp.</i>
AMCH131	36	M	MU-VI	LRTI	Amoxyclav	Sputum	<i>Enterobacter sp.</i>
AMCH133	5	M	PAED	UTI	Amoxyclav	Urine	<i>E. coli</i>
AMCH140	1	F	OBN	Newborn Sepsis	Cefotaxime Injection	Blood	<i>K. pneumoniae</i>
AMCH 142	15	M	Ortho-I	Surgery	Ceftriaxone	Wound swab	<i>P. aeruginosa</i>
AMCH146	50	M	MU-VII	Pneumoniae	Azithromycin	Tracheal Aspirate	<i>K. pneumoniae</i>
AMCH149	14	M	SU-I	Surgical Wound	Ceftriaxone	Pus	<i>K. pneumoniae</i>
AMCH150	2	F	OBN	Sepsis	Amoxyclav	Blood	<i>E. coli</i>
AMCH153	1	M	OBN	Sepsis	Amoxyclav	Blood	<i>E. coli</i>
AMCH155	1	M	OBN	Sepsis	Amoxyclav	Blood	<i>K. pneumoniae</i>
AMCH 159	1	F	OBN	Sepsis	Amoxyclav	Blood	<i>K. pneumoniae</i>
AMCH161	66	M	URO	Prostate Enlargement	Nitrofurantoin	Urine	<i>Citrobacter sp.</i>
AMCH174	55	M	OPD	Dysentery	Norfloxacin	Stool	<i>E. coli</i>
AMCH182	60	F	MU-VII	Pneumoniae	Azithromycin	Tracheal Aspirate	<i>P. aeruginosa</i>
AMCH 195	1	F	OBN	Sepsis	Ceftriaxone	Blood	<i>E. coli</i>
AMCH204	28	M	SU-I	Gangrene	Ceftriaxone	Pus	<i>A. baumannii</i>
AMCH206	43	F	SU-V	Surgical Wound	Cefixime	Pus	<i>A. baumannii</i>
AMCH209	55	M	OPD	Diarrhoea	Ofloxacin	Stool	<i>Salmonella sp.</i>
AMCH217	30	M	MU-VII	Pneumoniae	Amoxyclav + Azithromycin	Tracheal Aspirate	<i>A. baumannii</i>
AMCH218	65	M	MU-VII	Pneumoniae	Azithromycin	Tracheal Aspirate	<i>A. baumannii</i>
AMCH404	24	F	OPD	Abdominal Pain	Nitrofurantoin	Urine	<i>K. pneumoniae</i>
AMCH406	2	M	OBN	Sepsis	Cefotaxime Injection	Blood	<i>E. coli</i>
AMCH412	45	M	SU-I	Surgical Wound	Ceftriaxone	Pus	<i>K. pneumoniae</i>
AMCH424	63	M	OPD	Dysentery	Ciprofloxacin	Stool	<i>Shigella sp.</i>
AMCH467	2	M	OBN	Sepsis	Cefuroxime Injection	Blood	<i>Proteus sp.</i>
AMCH 480	45	F	SU-II	Surgical Wound	Cefotaxime	Pus	<i>Enterobacter sp.</i>
AMCH 508	62	M	OPD	Diarrhoea	Norfloxacin	Stool	<i>Shigella boydii</i>
AMCH509	45	F	OPD	Diarrhoea	Ofloxacin	Stool	<i>Shigella dysenteriae</i>
AMCH510	55	F	OPD	Diarrhoea	Ofloxacin	Stool	<i>Shigella dysenteriae</i>
AMCH511	18	F	OPD	Diarrhoea	Ofloxacin	Stool	<i>Shigella dysenteriae</i>
BPIIC007	25	F	OPD	Ante Natal Care	Amoxicillin	Urine	<i>E. coli</i>
BPIIC010	23	F	OPD	Ante Natal Care	Ampicillin	Blood	<i>E. coli</i>
UTI1	45	F	OPD	UTI	Nitrofurantoin	Urine	<i>E. coli</i>
UTI2	55	F	OPD	Abdominal Pain	Cefotaxime	Urine	<i>K. pneumoniae</i>

Table 3

Distribution of various pathogens obtained from clinical isolates.

Source of the organism	<i>E. coli</i> (%)	<i>Klebsiella sp.</i> (%)	<i>A. baumannii</i> (%)	<i>Pseudomonas sp.</i> (%)	<i>Citrobacter sp.</i> (%)	<i>Shigella sp.</i> (%)	<i>Enterobacter sp.</i> (%)	<i>Salmonella sp.</i> (%)	<i>Proteus sp.</i> (%)
Blood	13(28%)	7(28%)	4(31%)	2(22%)	1(20%)	0	0	0	100
Urine	20(43%)	10(40%)	1(7%)	0	4(80%)	0	0	0	0
Tracheal aspirate	1(2%)	2(8%)	5(38%)	5(33%)	0	0	0	0	0
Pus	0	2(8%)	3(23%)	0	0	0	2(66%)	0	0
Stool	9(19%)	1(4%)	0	0	0	5(100%)	0	2(100%)	0
Wound Swab	1(2%)	1(4%)	0	4(44%)	0	0	0	0	0
Sputum	3(6%)	2(8%)	0	0	0	0	1(33%)	0	0
Total	47 (42.7%)	25(22.7%)	13(11.8%)	9(8.18%)	5(4.54%)	5(4.54%)	3(2.7%)	2(1.8%)	1(0.9%)

method among which 42 (38%) were found to be only ESBL producers, 26 (24%) were only MBL producers and 12 (11%) were both ESBL and MBL producers. Among the total 80 β -lactamase producers two predominant organisms were *E. coli* and *K. pneumoniae* found 43.7% and 25% respectively (Table 5).

Characterization of β -lactamase producing genes by PCR

Total 80 isolates were tested positive for the production of β -lactamase enzyme either ESBL, or MBL and both and all these isolates were subjected for screening for the genes known to be responsible for the development of beta-lactamases. Among them, 78 isolates showed the presence of the genes by PCR assay. Among the total 54 (42 + 12) ESBL positive isolates, 29(53.7%) harbored *blaCTX-M* gene, 13 (24%) had *blaTEM*, 10 (18.5%) had *blaSHV*, and 9(16.6%) isolates had *blaOXA-1* gene. No isolates were tested positive for *blaOXA-7* (Table 5). All the *blaCTX-M* genes found in the ESBL positive isolates were confirmed to be *CTX-M3* type. The sequence has been submitted to NCBI GeneBank. Sequencing of all the 13

(24%) isolates positive for *blaTEM* gene revealed the presence of *blaTEM-1* type, which is not a ESBL.

Among 38 (26 + 12) MBL positive isolates, 18(47.3%) isolates were tested positive for *VIM-1*, followed by *blaNDM-1*(10, 26.3%), *IMP-1* (6; 15.7%) and 3 (7.8%) isolates positive for *OXA-48* (class D metallo beta lactamase) and 2 organisms one *E. coli* and *K. pneumoniae* each had *SIM-1* and *SPM-1*. *SIM-1* and *SPM-1* genes belong to class B MBL. No isolates were positive for *GIM-1*. The sequences of *SIM-1* and *SPM-1* were deposited in gene bank.

Determination of fluoroquinolone resistance by efflux pump assay

Among 110 isolates, 90 (81.8%) were resistant to nalidixic acid (NAL) and 50 (45.4%) were resistant to all five fluoroquinolones tested. Among these 50 isolates, only 15 (30%) showed the efflux pump mechanism for fluoroquinolone resistance, by reduction in the MIC 2–8 fold with the addition of CCCP along with ciprofloxacin in comparison to ciprofloxacin alone. All these 15 isolates had double mutations (S83L and D87N) in *gyrA* and single mutation (S80I) in *parC* (Table 6).

Table 4
Pattern of antibiotic resistance among the clinical pathogens.

Antimicrobial agent	<i>E. coli</i> (47)	<i>K. pneumoniae</i> (25)	<i>A. baumannii</i> (13)	<i>Citrobacter</i> sp. (5)	<i>P. aeruginosa</i> (9)	<i>Shigella</i> sp. (5)	<i>Enterobacter</i> sp. (3)	<i>Salmonella</i> sp. (2)	<i>Proteus</i> sp (1)	Total resistance
Ampicillin (AMP)	93.6(44)	92 (23)	92(12)	80(4)	89(8)	60(3)	100(3)	100(2)	100(1)	90.9 (100)
Amikacin (AMK)	31.9(15)	36(9)	61.5(8)	60(3)	44.4(4)	40(2)	66.6(2)	100(2)	0	40.9(45)
Gentamicin (GEN)	21.2(10)	24(6)	61.5(8)	40(2)	33.3(3)	40(2)	66.6(2)	100(2)	0	31.8(35)
Cefotaxime (CTX)	93.6(44)	84(21)	77(10)	80(4)	77.7(7)	60(3)	100(3)	50(1)	100(1)	85.4 (94)
Ceftazidime (CAZ)	93.6(44)	84(21)	77(10)	80(4)	77.7(7)	60(3)	100(3)	50(1)	100(1)	85.4 (94)
Ceftriaxone (CTR)	95.7(45)	92(23)	84.6(11)	80(4)	77.7(7)	80 (4)	100(3)	50(1)	100(1)	90.0 (99)
Ciprofloxacin (CIP)	42.5(20)	48(12)	38.4(5)	40(2)	44.4(4)	40(2)	100(3)	50(1)	100(1)	45.4 (50)
Norfloxacin (NOR)	42.5(20)	48(12)	38.4(5)	40(2)	44.4(4)	40(2)	100(3)	50(1)	100(1)	45.4 (50)
Gatifloxacin (GAT)	42.5(20)	48(12)	38.4(5)	40(2)	44.4(4)	40(2)	100(3)	50(1)	100(1)	45.4 (50)
Ofloxacin (OFX)	42.5(20)	48(12)	38.4(5)	40(2)	44.4(4)	40(2)	66.6(2)	50(1)	100(1)	45.4 (50)
Levofloxacin (LEV)	42.5(20)	48(12)	38.4(5)	40(2)	44.4(4)	40(2)	66.6(2)	50(1)	100(1)	45.4 (50)
Chloramphenicol (CHL)	53.1(25)	48(12)	38.4(5)	40(2)	33.3(3)	40(2)	66.6(2)	50(1)	100(1)	48.1(53)
Tetracycline (TET)	57.4(27)	52(13)	46.1(6)	60(3)	44.4(4)	60(3)	33.3(1)	50(1)	100(1)	53.6(59)
Piperacillin-tazobactam (PTZ)	25.5(12)	24(6)	61.5(8)	20(1)	33.3(3)	40(2)	66.6(2)	50(1)	100(1)	32.7(36)
Imipenem (IPM)	25.5(12)	36(9)	46.1(6)	40(2)	33.3(3)	40(2)	66.6(2)	50(1)	100(1)	34.5 (38)
Cefotaxime-clavulanic acid (CEC)	27.6(13)	24(6)	61.5(8)	20(1)	33.3(3)	40(2)	66.6(2)	50(1)	100(1)	33.6(37)
Nitrofurantoin (NIT)	38.2(18)	44(11)	38.4(5)	60(3)	33.3(3)	40(2)	66.6(2)	50(1)	100(1)	41.8(46)
Cefepime (CFM)	47.1(20)	40(10)	38.4(5)	60(3)	33.3(3)	40(2)	66.6(2)	50(1)	100(1)	42.7(47)
Nalidixic acid (NAL)	89.3(42)	72(18)	77(10)	80(4)	77.7(7)	60(3)	100(3)	100(2)	100(1)	81.8(90)
Cotrimethoxazole (CoT)	78.7(37)	60(15)	53.8(7)	80(4)	44.4(4)	40(2)	66.6(2)	50(1)	100(1)	66.3(73)

Table 5
Distribution of ESBL and MBL genes in among the clinical pathogens.

SI No	Strain no.	Organism	MIC (μg/mL)				ESBL/MBL	ESBL Genes	MBL Genes
			Cefotaxime	Ceftazidime	Imipenem	Meropenem			
1	AMCH 1	<i>A. baumannii</i>	1024	1024	0.5	0.5	+/-	CTX-M	-
2	AMCH 3	<i>A. baumannii</i>	4096	4096	256	128	-/+	-	VIM-1
3	AMCH 4	<i>E. coli</i>	4096	4096	64	64	-/+	-	NDM-1
4	AMCH 6	<i>A. baumannii</i>	2048	1024	64	64	+/+	CTX-M	VIM-1
5	AMCH 7	<i>K. pneumoniae</i>	1024	1024	0.5	0.5	+/-	TEM	-
6	AMCH 8	<i>E. coli</i>	4096	4096	32	32	+/+	TEM	OXA-48
7	AMCH 9	<i>A. baumannii</i>	4096	4096	0.25	0.12	+/-	SHV	-
8	AMCH 10	<i>E. coli</i>	4096	4096	128	128	-/+	-	VIM-1
9	AMCH 11	<i>E. coli</i>	1024	1024	0.12	0.25	+/-	OXA1	-
10	AMCH 13	<i>Citrobacter</i> sp.	256	1024	0.12	0.12	+/-	OXA1, SHV	-
11	AMCH 16	<i>K. pneumoniae</i>	2048	1024	128	256	+/+	TEM	SIM-1
12	AMCH 17	<i>P. aeruginosa</i>	2048	2048	0.12	0.12	+/-	TEM	-
13	AMCH 18	<i>E. coli</i>	512	1024	128	256	-/+	-	OXA-48
14	AMCH 25	<i>K. pneumoniae</i>	256	512	128	128	-/+	-	NDM-1
15	AMCH 26	<i>E. coli</i>	1024	1024	0.12	0.25	-/+	-	VIM-1
16	AMCH 27	<i>E. coli</i>	1024	1024	0.12	0.25	+/-	CTX-M	-
17	AMCH 28	<i>P. aeruginosa</i>	1024	1024	0.25	0.5	+/-	SHV	-
18	AMCH 29	<i>A. baumannii</i>	2048	1024	0.12	0.25	+/+	CTX-M	-
19	AMCH 31	<i>E. coli</i>	2048	2048	2	4	+/-	SHV, CTX-M3	-
20	AMCH 32	<i>K. pneumoniae</i>	1024	1024	1	2	+/-	TEM	-
21	AMCH 33	<i>E. coli</i>	2048	1024	0.5	0.5	-	CTX-M	-
22	AMCH 34	<i>E. coli</i>	1024	1024	0.5	0.5	+/-	CTX-M3	-
23	AMCH 35	<i>A. baumannii</i>	1024	512	1	1	+/-	TEM	-
24	AMCH 36	<i>P. aeruginosa</i>	512	1024	0.5	1	+/-	CTX-M	-
25	AMCH 38	<i>P. aeruginosa</i>	1024	512	1	1	+/-	CTX-M	-
26	AMCH 39	<i>E. coli</i>	2048	2048	1	2	+/-	TEM	-
27	AMCH 42	<i>E. coli</i>	1024	2048	128	256	-/+	-	VIM-1
28	AMCH 44	<i>K. pneumoniae</i>	512	1024	64	64	+/+	CTX-M	VIM-1
29	AMCH 47	<i>K. pneumoniae</i>	1024	1024	1	0.5	+/-	SHV	-
30	AMCH 48	<i>K. pneumoniae</i>	2048	1024	1	0.5	+/-	SHV	-
31	AMCH 50	<i>P. aeruginosa</i>	256	512	128	256	-/+	-	VIM-1
32	AMCH 53	<i>E. coli</i>	512	1024	0.25	0.25	+/-	TEM	-
33	AMCH 54	<i>E. coli</i>	1024	1024	128	128	+/+	SHV	VIM-1
34	AMCH 55	<i>E. coli</i>	2048	4096	1	2	+/-	CTX-M3, OXA-1	-
35	AMCH 56	<i>E. coli</i>	1024	2048	1	1	+/-	SHV	-
36	AMCH 57	<i>Citrobacter</i> sp.	2048	1024	128	256	+/+	TEM	NDM-1
37	AMCH 59	<i>E. coli</i>	1024	1024	0.5	0.25	+/-	TEM, OXA-1	-
38	AMCH 60	<i>A. baumannii</i>	2048	4096	256	256	-/+	-	NDM-1
39	AMCH 61	<i>K. pneumoniae</i>	512	1024	256	512	-/+	-	OXA-48
40	AMCH 62	<i>E. coli</i>	1024	2048	0.5	0.125	+/-	TEM	-
41	AMCH 64	<i>K. pneumoniae</i>	2048	1024	0.5	0.5	+/-	TEM, OXA-1	-
42	AMCH 68	<i>K. pneumoniae</i>	1024	1024	0.5	0.5	+/-	CTX-M	-

Table 5 (Continued)

SI No	Strain no.	Organism	MIC (μg/mL)				ESBL/MBL	ESBL Genes	MBL Genes
			Cefotaxime	Ceftazidime	Imipenem	Merpenem			
43	AMCH 69	<i>E. coli</i>	2048	4096	128	128	-/+	–	VIM-1
44	AMCH 73	<i>E. coli</i>	4096	2048	128	256	-/+	–	VIM-1
45	AMCH 74	<i>E. coli</i>	2048	2048	256	256	-/+	–	VIM-1
46	AMCH 78	<i>P. aeruginosa</i>	1024	2048	128	128	+/+	CTX-M	IMP-1
47	AMCH 84	<i>E. coli</i>	4096	4096	512	256	-/+	–	NDM-1,SPM-1
48	AMCH 85	<i>E. coli</i>	2048	4096	256	512	-/+	–	IMP-1
49	AMCH 86	<i>K. pneumoniae</i>	1024	1024	256	256	-/+	–	IMP-1
50	AMCH 87	<i>K. pneumoniae</i>	4096	4096	128	256	+/+	CTX-M	VIM-1
51	AMCH 89	<i>K. pneumoniae</i>	2048	4096	0.5	0.5	+/-	SHV	–
52	AMCH 92	<i>E. coli</i>	1024	2048	1	2	+/-	CTX-M3	–
53	AMCH 93	<i>E. coli</i>	4096	4096	256	256	-/+	–	VIM-1
54	AMCH 96	<i>K. pneumoniae</i>	2048	2048	512	512	-/+	–	IMP-1
55	AMCH 97	<i>E. coli</i>	4096	4096	128	256	+/+	OXA-1	VIM-1
56	AMCH 104	<i>E. coli</i>	4096	4096	64	128	+/+	CTX-M	NDM-1
57	AMCH 106	<i>E. coli</i>	128	256	1	2	+/-	SHV	–
58	AMCH 131	<i>Enterobacter sp.</i>	4096	4096	2	4	+/-	CTXM-3, OXA-1	–
59	AMCH 133	<i>E. coli</i>	1024	1024	1	2	+/-	OXA-1	–
60	AMCH 140	<i>K. pneumoniae</i>	2048	2048	0.25	0.5	+/-	TEM	–
61	AMCH 142	<i>P. aeruginosa</i>	1024	1024	1	2	+/-	CTX-M3	–
62	AMCH 146	<i>K. pneumoniae</i>	1024	2048	0.5	0.5	+/-	CTX-M	–
63	AMCH 153	<i>E. coli</i>	8192	4096	64	64	+/-	CTX-M	–
64	AMCH 155	<i>K. pneumoniae</i>	1024	2048	256	256	+/+	CTX-M	VIM-1
65	AMCH159	<i>K. pneumoniae</i>	4096	4096	0.12	0.12	+/-	CTX-M	–
66	AMCH 161	<i>Citrobacter sp.</i>	2048	4096	128	128	+/+	CTX-M	VIM-1
67	AMCH174	<i>E. coli</i>	4096	4096	0.12	0.12	+/-	CTX-M	–
68	AMCH 195	<i>E. coli</i>	8192	4096	64	64	+/-	CTX-M	–
69	AMCH 204	<i>A. baumannii</i>	4096	4096	128	256	-/+	–	NDM-1
70	AMCH 217	<i>A. baumannii</i>	4096	4096	256	512	-/+	–	VIM-1
71	AMCH 218	<i>A. baumannii</i>	2048	2048	256	256	-/+	–	NDM-1
72	AMCH 404	<i>K. pneumoniae</i>	4096	4096	128	128	-/+	–	VIM-1
73	AMCH 406	<i>E. coli</i>	8192	4096	4	4	+/-	CTX-M	–
74	AMCH 467	<i>Proteus sp.</i>	2048	2048	512	512	-/+	–	NDM-1
75	AMCH 480	<i>Enterobacter sp.</i>	2048	4096	256	256	-/+	–	NDM-1
76	AMCH 510	<i>Shigella dysenteriae</i>	2048	2048	256	256	-/+	–	IMP-1
77	AMCH 511	<i>Shigella boydii</i>	2048	2048	128	256	-/+	–	IMP-1
78	BPIIC 007	<i>E. coli</i>	4096	2048	0.5	0.5	+/-	CTX-M3, OXA-1	–
79	UTI1	<i>E. coli</i>	8192	4096	1	2	+/-	CTX-M	–
80	UTI 2	<i>K. pneumoniae</i>	2048	2048	0.12	0.25	+/-	CTX-M	–

Determination of fluoroquinolone resistance by mutation in QRDR region

The QRDRs regions of the twenty isolates revealed the presence of double mutations in *GyrA* (S83L and D87N) where serine(S) was replaced with leucine (L) at the codon position 83 and aspartic acid (D) was replaced with asparagine(N) at the position 87.

Nineteen isolates had single mutation (S80I) at the codon position 80 of *parC* with a replacement of serine (S) to isoleucine(I) and one isolate showed (E84V) the presence of valine (V) in replacement of glutamic acid (E) at the codon position 84 (Table 6).

Presence of PMQR determinants in fluoroquinolone resistant strains

Among the 20 isolates which were screened for the presence of PMQR regions, 12 isolates were positive for *qnrA* gene and 3 isolates showed the presence of *qnrS* gene and 2 of them showed the presence of both genes *qnrA* and *qnrS*. But among them only 5 isolates harbored the presence of *aac(6')-Ib-cr* gene and no isolates were tested positive for *QepA* gene.

Among the isolates tested positive for PMQR, all had double mutations (S83L and D87N) in *gyrA* and single mutation (S80I) in *parC* (Table 6).

Discussion

Epidemiological studies are important to understand the antimicrobial resistance burden in hospital and community and

to provide correct medication and also to develop a strategy to combat multidrug resistance [24]. Gram negative bacteria are commonly found associated with health care infection and causes more mortality than gram positive bacteria [25]. In the present study we have found that gram negative bacteria were isolated mostly from urine (32%) followed by blood (25%). Other studies also supported that urinary tract infection (UTI) infections were the most common cause for patients coming or getting admitted in hospital [8].

Urinary tract infections are mostly caused by *E. coli*. This study reported the presence of *E. coli* (43%) and *K. pneumoniae* (23%) as the two most common pathogens, followed by *A. baumannii* (12%) and *P. aeruginosa* (8%). Previous studies also reported the high prevalence of *E. coli* and *K. pneumoniae* in healthcare setting [27].

In case of infection caused by gram negative bacteria, β -lactams and fluoroquinolones are generally prescribed because of their broad spectrum activity and effectiveness. However irrational use of antibiotics lead bacteria to an edge to develop various resistance mechanisms against them and create “nightmare” for mankind. The present study also demonstrates that the strains isolated from various clinical cases are having high resistance against all commonly used antibiotics like ampicillin, all 3rd gen cephalosporins, and fluoroquinolones. Least resistance was observed against amikacin, gentamicin and imipenem which is similar to other report from this area [28].

Emerging prevalence of multidrug resistant bacteria infection in community and hospital acquired infection is a concerning matter for North East India. Generally, all over the world, the rising causes of resistance are limited knowledge about antibiotics, less guidance, self-medication and ignorance [28]. In our study,

Table 6
Details of MIC of fluoroquinolones, quinolone resistance-determining region (QRDR) mutations, plasmid-mediated quinolone resistance (PMQR) determinants and prevalence of the antibiotic resistance conferring genes among the clinical isolates.

Sl. no	Strain	Organism	Isolation source	MIC ($\mu\text{g/mL}$)		OFX	LEV	GAT	MIC OF CIP + CCCP ($\mu\text{g/mL}$)	gyrA mutation		parC mutation		PMQR		aac(6')-Ib-cr
				CIP	NOR					S83L	D87N	S80I	E84V	qnrA	qnrS	
1	AMCH 1	<i>A. baumannii</i>	Blood	128	128	128	64	64	16	+	+	+	-	-	-	-
2	AMCH 2	<i>E. coli</i>	Urine	>256	128	128	128	128	64	+	+	+	-	+	-	-
3	AMCH 4	<i>E. coli</i>	Blood	128	128	64	64	64	8	+	+	+	+	+	-	-
4	AMCH 5	<i>K. pneumoniae</i>	Blood	32	32	32	32	32	4	+	+	+	-	-	+	-
5	AMCH 26	<i>E. coli</i>	Sputum	128	128	128	64	64	16	+	+	+	-	+	+	-
6	AMCH 27	<i>E. coli</i>	Urine	64	64	64	32	32	8	+	+	+	-	+	-	+
7	AMCH 56	<i>E. coli</i>	Urine	64	64	64	32	32	4	+	+	+	-	+	-	-
8	AMCH 59	<i>E. coli</i>	Urine	128	64	64	32	32	16	+	+	+	-	+	-	+
9	AMCH 60	<i>A. baumannii</i>	Blood	128	128	256	128	64	32	+	+	+	-	-	+	+
10	AMCH 69	<i>E. coli</i>	Stool	256	>256	256	128	128	32	+	+	+	-	-	-	-
11	AMCH 80	<i>E. coli</i>	Urine	32	32	32	16	16	4	+	+	+	-	+	-	-
12	AMCH 84	<i>E. coli</i>	Blood	256	128	128	128	128	64	+	+	+	-	+	+	-
13	AMCH 85	<i>E. coli</i>	Blood	64	64	32	32	32	16	+	+	+	-	-	-	-
14	AMCH 86	<i>K. pneumoniae</i>	Urine	64	64	64	32	32	16	+	+	+	-	+	+	+
15	AMCH 96	<i>K. pneumoniae</i>	Urine	64	64	64	16	16	8	+	+	+	-	-	-	-
16	AMCH 106	<i>E. coli</i>	Urine	64	64	32	32	8	4	+	+	+	-	-	-	-
17	AMCH 108	<i>E. coli</i>	Urine	128	256	64	64	64	32	+	+	+	-	+	-	-
18	AMCH 161	<i>Citrobacter sp.</i>	Urine	>256	256	128	128	128	32	+	+	+	-	-	-	-
19	AMCH 480	<i>Enterobacter sp.</i>	Pus	>256	>256	256	256	256	64	+	+	+	-	+	-	-
20	BPIC007	<i>E. coli</i>	Urine	256	128	256	128	128	64	+	+	+	-	+	-	-

the prevalence of MDR bacteria was 77.2% which is much higher than 67.6% reported by Sharma et al., 2018 from north east India [29,30]. Beside irresponsible usage of antibiotics by human, an unknown continuous exposure of antibiotic is happening, which all are cumulatively contributing toward the steep gradient of antibiotic resistance [26].

Increased ESBL production is linked to intake of carbapenem for treatment, which further develop carbapenem resistance [15]. The present study reported to have 24% MBL producers among total isolates and 11% bacteria possess both ESBL and MBL genes. Devi et al., also reported the increased prevalence of carbapenemase gene in hospital based infections even in neonates [31]. *E. coli* and *K. pneumoniae* were detected as highest ESBL and MBL producers followed by *A. baumannii* and *P. aeruginosa*. *Citrobacter* sp. was found to possess ESBL gene in three isolates and two isolates have both ESBL and MBL gene, which was not reported earlier from this region.

However, analyzing genes responsible for the production of β -lactamase enzymes, 93% (n = 67) of the bacteria showed the presence of genes. For ESBL enzymes, *blaCTX-M3* (52%) was found in majority followed by *blaSHV* (18.5%). None of the strains were found to be positive for *blaOXA-7* and *blaOXA-1* was mostly present either in combination with other ESBL genes or MBL genes. Amani et al., (2020) reported to have the same order of prevalence of beta lactamase genes in isolates and also reported to have combinations of ESBL and MBL genes [32].

The present study found the increased prevalence of *VIM-1* (44.7%) MBL gene, followed by *blaNDM-1* (23.6%) gene and *IMP-1* (15.7%). Three isolates were reported to have *OXA-48* gene. As reported earlier, *NDM-1* gene is not predominant in this geographical region, like North India [31,33]. *SIM-1* (Seoul imipenemas) was initially identified in Korea in 2005 and *SPM-1* (Sao Paulo metallo-beta-lactamase) gene was detected in Brazil in 1997. *VIM-1* and *IMP-1* are the most widely spread MBL genes found carbapenem resistant bacteria [34]. Our study, perhaps for the first time in India, also establishes the presence of the *SIM-1* and *SPM-1* in *E. coli* and *Klebsiella* sp.

In the present study, 45.4% of the strains were resistant to all five fluoroquinolones tested.

Vertical inheritance of spontaneous point mutations in GyrA and ParC are considered as the primary cause of fluoroquinolone resistance [35]. The present study also shows that two mutations in *gyrA* (S83L and D87N) and one mutation in *parC* (S80I) are commonly linked to fluoroquinolone resistance. Another mutation detected in *parC* region, which resulted in replacement of valine (V) to glutamic acid (E) at amino acid position 84, which has also reported in *E. coli* earlier and is responsible for disfunctioning of topoisomerase (*parC*) enzyme [36].

Along with mutations in QRDR regions, the presence of PMQR determinants is a matter of concern in regards to fluoroquinolone resistance, because plasmid mediated resistant genes can be horizontally transferred among other bacteria which further increase the spread of resistance [18]. In the present study, among 20 tested isolates, high prevalence of *qnrA* (60%) was found than *qnrS*. Salah et al., also reported the concomitant presence of both *qnrA* and *qnrS* [36]. Another plasmid mediated quinolone resistant gene, *aac(6')-Ib-cr* is responsible for fluoroquinolone resistance by modifying fluoroquinolones with piperazinyl moiety [37]. Our study reported the presence of *aac(6')-Ib-cr* in 5 strains. The current study indicates that, along with multiple mutations in QRDR regions, perhaps the presence of PMQR plays an important role in increasing the resistance towards fluoroquinolones.

The present study reported the decrease of two to fourfold in MIC when uncoupling agent CCCP was added in MHA and the presence of efflux pump is found to be another prominent reason of fluoroquinolone resistance [39]. Among the various resistance mechanisms, efflux pump is proven to have important role in

antibiotic resistance specially in structurally unrelated antibiotics, beta-lactam, TET, MTZ and CIP [40].

Conclusion

Detailed knowledge of the infection causing bacteria, antibiotic susceptibility profile, and knowledge about the exact cause of resistance, geographical pattern of resistance are the important factors for developing correct healthcare guideline to combat AMR resistance and also which will eventually reduce the mortality rate. The study represents the first presentation of the results from NE India covering the priority pathogens isolated from hospital and community based infection. The present study showed the increased prevalence of ESBL, MBL producers and also increased fluoroquinolone resistance among the isolates. Similar like other reports from different parts of the world, this study also confirms the emergence of multi drug resistance in North East India from hospital and community based infections. This study also confirmed the emergence of new MBL gene (*SIM-1* and *SPM-1*) in clinical isolates. The informations gathered during this study could be useful for development of new treatment strategies for the community and also provide baseline data which could help to employ antibiotic policy to reduce the spread of drug resistance.

Accession Number of NCBI Genbank Submission

The sequences in this study have been deposited in the GenBank database under accession numbers MT872795 (*SIM-1*), MT872796 (*SPM-1*), GenBank BankIt ID 2405631 (*blaCTX-M*) and GenBank BankIt ID 2405616 (*blaCTX-M3*).

Ethics approval

The study was cleared by institutional ethical committee.

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Transparency declaration

None to declare.

Conflict of interests

The authors declare that they have no conflict of interest.

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